# Useful Identities

We will prove a few identities used by Jackson, whose proofs are not provided.

## 1 $\int_V \nabla\psi \; d^3x = \int_S \psi\mathbf{n} \; da$

To prove this identity, notice that

$$\mathbf{c}\cdot\nabla\psi = \nabla\cdot(\psi\mathbf{c}) - \psi\nabla\cdot\mathbf{c} = \nabla\cdot(\psi\mathbf{c})\,,$$

for an arbitrary constant vector $\mathbf{c}$. Then, applying Gauss theorem, we have

$$\mathbf{c}\cdot\int_V \nabla\psi \; d^3x = \int_V \nabla\cdot(\psi\mathbf{c}) \; d^3x = \int_S (\psi\mathbf{c})\cdot\mathbf{n} \; da = \mathbf{c}\cdot\int_S \psi\mathbf{n} \; da.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_V \nabla\psi \; d^3x = \int_S \psi\mathbf{n} \; da.$$

This identity is used to transform Eq. (4.14) to Eq. (4.15).

## 2 $\int_V \nabla\times\mathbf{A} \; d^3x = \int_S \mathbf{n}\times\mathbf{A} \; da$

Again, for arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c}\cdot(\nabla\times\mathbf{A}) = \nabla\cdot(\mathbf{A}\times\mathbf{c}) + \mathbf{A}\cdot(\nabla\times\mathbf{c}) = \nabla\cdot(\mathbf{A}\times\mathbf{c})\,.$$

Applying Gauss theorem,

$$\mathbf{c}\cdot\int_V \nabla\times\mathbf{A} \; d^3x = \int_V \nabla\cdot(\mathbf{A}\times\mathbf{c}) \; d^3x = \int_S \mathbf{n}\cdot(\mathbf{A}\times\mathbf{c}) \; da = \mathbf{c}\cdot\int_S (\mathbf{n}\times\mathbf{A}) \; da.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_V \nabla\times\mathbf{A} \; d^3x = \int_S \mathbf{n}\times\mathbf{A} \; da.$$

This identity is applied to Eq. (5.60) to convert the integration over the entire space to a surface integral.

## 3 $\int_S \mathbf{n} \times \nabla\psi \; da = \oint_C \psi \; d\mathbf{l}$

Again, for arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c} \cdot (\mathbf{n} \times \nabla\psi) = \mathbf{n} \cdot (\nabla\psi \times \mathbf{c}) = \mathbf{n} \cdot [\nabla \times (\psi\mathbf{c}) - \psi\,(\nabla \times \mathbf{c})] = \mathbf{n} \cdot [\nabla \times (\psi\mathbf{c})]\,.$$

Applying Stokes theorem,

$$\mathbf{c} \cdot \int_S (\mathbf{n} \times \nabla\psi) \;\; da = \int_S \mathbf{n} \cdot [\nabla \times (\psi\mathbf{c})] \;\; da = \oint_C (\psi\mathbf{c}) \cdot \;\; d\mathbf{l} = \mathbf{c} \cdot \oint_C \psi \; d\mathbf{l}.$$

Since $\mathbf{c}$ is arbitrary, we must have

$$\int_S \mathbf{n} \times \nabla\psi \; da = \oint_C \psi \; d\mathbf{l}.$$

## 4 $\oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A})\,\mathbf{n} \; da - \int_S (\nabla\mathbf{A}) \cdot \mathbf{n} \; da$

For arbitrary constant vector $\mathbf{c}$,

$$\mathbf{c} \cdot \oint_C \mathbf{A} \times d\mathbf{l} = \oint_C (\mathbf{c} \times \mathbf{A}) \cdot d\mathbf{l} = \int_S [\nabla \times (\mathbf{c} \times \mathbf{A})] \cdot \mathbf{n} \; da.$$

Since

$$\nabla \times (\mathbf{c} \times \mathbf{A}) = \mathbf{c}\,(\nabla \cdot \mathbf{A}) - \mathbf{A}\,(\nabla \cdot \mathbf{c}) + (\mathbf{A} \cdot \nabla)\,\mathbf{c} - (\mathbf{c} \cdot \nabla)\,\mathbf{A} = \mathbf{c}\,(\nabla \cdot \mathbf{A}) - (\mathbf{c} \cdot \nabla)\,\mathbf{A},$$

we will have

$$\mathbf{c} \cdot \oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A})\,(\mathbf{c} \cdot \mathbf{n}) \;\; da - \int_S (\mathbf{c} \cdot (\nabla\mathbf{A}) \cdot \mathbf{n}) \;\; da = \mathbf{c} \cdot \left[\int_S (\nabla \cdot \mathbf{A})\,\mathbf{n} \; da - \int_S (\nabla\mathbf{A}) \cdot \mathbf{n} \; da\right],$$

or

$$\oint_C \mathbf{A} \times d\mathbf{l} = \int_S (\nabla \cdot \mathbf{A})\,\mathbf{n} \; da - \int_S (\nabla\mathbf{A}) \cdot \mathbf{n} \; da.$$

In particular, for $\mathbf{A} = \mathbf{x}$, we know

$$\nabla \cdot \mathbf{x} = 3, \quad (\nabla\mathbf{x}) \cdot \mathbf{n} = \sum_i \sum_j \hat{e}_j \frac{\partial x_i}{\partial x_j} n_i = \mathbf{n}.$$

Then,

$$\oint_C \mathbf{x} \times d\mathbf{l} = \int_S [3\mathbf{n} - \mathbf{n}] \;\; da = \int_S 2\mathbf{n} \; da = 2S\mathbf{n},$$

where $S$ is the area of the surface. This established Eq. (5.57).

## 5 $\int_V \mathbf{A}\ d^3x = -\int_V \mathbf{x}\left(\nabla\cdot\mathbf{A}\right)\ d^3x$

Consider the following obvious identity,

$$\nabla\cdot\left(x_i\mathbf{A}\right) = \left(\nabla x_i\right)\cdot\mathbf{A} + x_i\left(\nabla\cdot\mathbf{A}\right) = A_i + x_i\left(\nabla\cdot\mathbf{A}\right).$$

Then,

$$\sum_{i=1}^{3}\hat{e}_i\left[\nabla\cdot\left(x_i\mathbf{A}\right)\right] = \mathbf{A} + \mathbf{x}\left(\nabla\cdot\mathbf{A}\right).$$

Integrate over the entire space and apply the Guass theorem, the left hand side becomes

$$\sum_{i=1}^{3}\hat{e}_i\int_V \nabla\cdot\left(x_i\mathbf{A}\right)\ d^3x = \sum_{i=1}^{3}\hat{e}_i\int_S \mathbf{n}\cdot\left(x_i\mathbf{A}\right)\ da.$$

If $\mathbf{A}$ is localized and drops to zero at infinity, or decreases fast enough, $|\mathbf{A}| = o(1/r^3)$ for $r\to\infty$, the surface integral will be zero, which leaves

$$\int_V \left[\mathbf{A} + \mathbf{x}\left(\nabla\cdot\mathbf{A}\right)\right]\ d^3x = 0,$$

or,

$$\int_V \mathbf{A}\ d^3x = -\int_V \mathbf{x}\left(\nabla\cdot\mathbf{A}\right)\ d^3x$$

## 6 $\det\left(\delta_{ij} + a_i b_j\right) = 1 + \mathbf{a}\cdot\mathbf{b}$

This identity is not mentioned in Jackson, but can be useful for Prob. 6.2.

Notice that,

$$\begin{pmatrix} F & C \\ B & D \end{pmatrix} = \begin{pmatrix} I_n & C \\ 0 & D \end{pmatrix}\begin{pmatrix} F - CD^{-1}B & C \\ d^{-1}B & I_m \end{pmatrix} = \begin{pmatrix} F & 0 \\ B & I_m \end{pmatrix}\begin{pmatrix} I_n & F^{-1}C \\ 0 & D - BF^{-1}C \end{pmatrix},$$

where $F$ is an invertible $n\times n$ matrix, $B$ an $m\times n$ matrix, $C$ an $n\times m$ matrix, $D$ an invertible $m\times m$ matrix, and $I_k$ a $k\times k$ identity matrix. Then,

$$\det\begin{pmatrix} F & C \\ B & D \end{pmatrix} = \det\begin{pmatrix} I_n & C \\ 0 & D \end{pmatrix}\cdot\det\begin{pmatrix} F - CD^{-1}B & C \\ d^{-1}B & I_m \end{pmatrix} = \det D\ \det\left(F - CD^{-1}B\right),$$

and

$$\det\begin{pmatrix} F & C \\ B & D \end{pmatrix} = \det F\ \det\left(D - BF^{-1}C\right).$$

Now, let

$$F = I_n,\ D = I_1 \equiv 1,\ B = (b_1,\cdots,b_n),\ C = -(a_1,\cdots,a_n)^{\top},$$

then we have

$$\det\begin{pmatrix} I_n & C \\ B & I_1 \end{pmatrix} = \det I_n\ \det\left(1 - BC\right) = 1 + \sum_i a_i b_i,$$

and also

$$\det\begin{pmatrix} I_n & C \\ B & I_1 \end{pmatrix} = \det I_1\ \det\left(I_n - CB\right) = \det\left(\delta_{ij} + a_i b_j\right).$$

Therefore, we have established

$$\det\left(\delta_{ij} + a_i b_j\right) = 1 + \mathbf{a}\cdot\mathbf{b}.$$